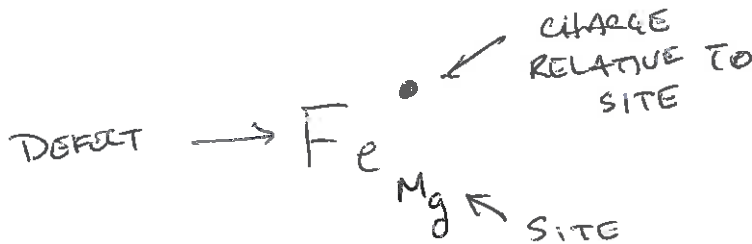


SOLID PHASE CONDUCTION

$$\sigma_i = n_i q_i \mu_i$$

DEFECT 'i' $\uparrow$ n_i CONCENTRATION
 CHARGE CARRIER $\uparrow$ q_i
 CHARGE $\uparrow$ μ_i MOBILITY

KRÖGER-VINK NOTATION



CHARGE	
-3	///
-2	//
-1	/
0	x
1	•
2	••
3	•••

$$n_i = N_i \exp\left(-\frac{\Delta H_{n_i}}{k_B T}\right)$$

MAXIMUM CONCENTRATION $\uparrow$ N_i
 BOLTZMANN'S CONSTANT $\uparrow$ k_B
 TEMP. IN K $\uparrow$ T
 ACTIVATION ENTHALPY $\uparrow$ ΔH_{n_i}

$$\mu_i = \mu_{i,0} \exp\left(-\frac{\Delta H_{\mu_i}}{k_B T}\right)$$

$$\begin{aligned} \sigma_i &= N_i \exp\left(-\frac{\Delta H_{n_i}}{k_B T}\right) q_i \mu_{i,0} \exp\left(-\frac{\Delta H_{\mu_i}}{k_B T}\right) \\ &= \underbrace{(N_i q_i \mu_{i,0})}_{\sigma_{i,0}} \exp\left(-\frac{(\Delta H_{n_i} + \Delta H_{\mu_i})}{k_B T}\right) \end{aligned}$$

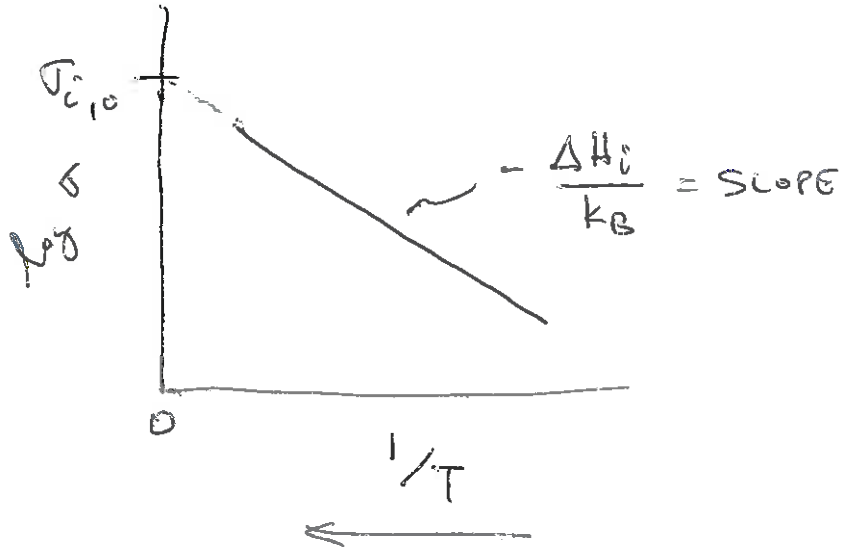
ΔH_i

$$\sigma_i = \sigma_{i,0} \exp\left(-\frac{\Delta H_i}{k_B T}\right)$$

$$\log_{10} \sigma_i = \log_{10} \left(\sigma_{i,0} \exp \left(- \frac{\Delta H_i}{k_B T} \right) \right)$$

$$\nearrow \text{WATERMAN LOG} = \log_{10} \sigma_{i,0} - \frac{\Delta H_i}{k_B T}$$

$$y(T) = a + m \frac{1}{T} = \text{LINE } \frac{1}{T}$$



MAXWELL'S EQUATIONS

$$\nabla \cdot \vec{E} = \frac{\rho_f}{\epsilon_0} \quad \text{GAUSS'S LAW OF ELECTRICITY}$$

$$\nabla \cdot \vec{H} = 0 \quad \text{GAUSS'S LAW OF MAGNETISM}$$

$$\nabla \times \vec{E} = -\mu_0 \frac{\partial \vec{H}}{\partial t} \quad \text{FARADAY'S LAW}$$

$$\nabla \times \vec{H} = \vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t} \quad \text{AMPERE'S LAW}$$

$$\vec{J} = \sigma \vec{E} \quad \text{CONSTITUTIVE RELATIONSHIP}$$

$$\nabla \times \vec{H} = \sigma \vec{E} + \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

FROM AMPERE'S LAW

$$\nabla \times (\nabla \times \vec{E}) = \nabla \times \left(-\mu_0 \frac{\partial \vec{H}}{\partial t} \right)$$

$$= -\mu_0 \frac{\partial}{\partial t} (\nabla \times \vec{H})$$

$$= -\mu_0 \frac{\partial}{\partial t} \left(\sigma \vec{E} + \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right)$$

VECTOR IDENTITY $\nabla \times \nabla \times \vec{A} = \nabla (\nabla \cdot \vec{A}) - \nabla^2 \vec{A}$

$$\nabla \cdot \vec{E} = \frac{\rho_f}{\epsilon_0} = 0$$

$$\nabla^2 \vec{E} = \mu_0 \sigma \frac{\partial \vec{E}}{\partial t} + \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} \quad \text{WAVE EQN.}$$

QUASISTATIC APPROXIMATION

$$\sigma \ll \omega \epsilon_0$$

↑
ANGULAR FREQUENCY

$$\nabla^2 \vec{E} = \mu_0 \sigma \frac{\partial \vec{E}}{\partial t} \quad \text{DIFFUSION EQN.}$$

$$\vec{E}(x, y, z, t) \xleftrightarrow{\mathcal{F}} \tilde{E}(x, y, z, \omega)$$

time-domain frequency-domain

1-D ASSUMPTION

DIFF. EQ. $\frac{\partial^2 \tilde{E}}{\partial z^2} = -k_z^2 \tilde{E}$

$$\frac{\partial^2 \tilde{E}}{\partial z^2} + k_z^2 \tilde{E} = 0$$

SOLN:

$$\tilde{E} = \underbrace{E^{(+)}(k_z, \omega) \exp(i k_z z)}_{\text{UPGOING}} + \underbrace{E^{(-)}(k_z, \omega) \exp(-i k_z z)}_{\text{DOWNGOING}}$$

$$k_z = (1 - i) \sqrt{\frac{\sigma \mu_0 \omega}{2}}$$

↑
COMPLEX # $i = \sqrt{-1}$

SKIN DEPTH, argument of exp

$$E = E_0 \exp(-1)$$

$$\arg = -1$$

$$\delta = \left(\frac{2}{\sigma \mu_0 \omega} \right)^{1/2} = [m]$$

$$\tilde{E} = Z \tilde{H}$$

↑

IMPEDANCE

$$Z = \frac{\tilde{E}_x}{\tilde{H}_y} = - \frac{E_y}{H_x}$$

$$\begin{bmatrix} E_x \\ E_y \end{bmatrix} = \begin{bmatrix} Z_{xx} & Z_{xy} \\ Z_{yx} & Z_{yy} \end{bmatrix} \begin{bmatrix} H_x \\ H_y \end{bmatrix}$$

$$E_x = \cancel{Z_{xx} H_x^0} + Z_{xy} H_y^0$$

$$E_y = Z_{yx} H_x + \cancel{Z_{yy} H_y^0}$$

1-D $Z_{xx} = 0 = Z_{yy}$ $Z_{xy} = -Z_{yx}$

$$2-D \quad z_{xy} \neq -z_{yx} \quad z_{xx} = z_{yy} = 0$$

$$3-D \quad z_{xy} \neq -z_{yx} \quad z_{xx} \neq 0, z_{yy} \neq 0$$

$$E_x = E_0 e^{-ik_z z}$$

$$H_y = H_0 e^{-ik_z z} = \frac{k_z}{\omega \mu_0} E_0 e^{-ik_z z}$$

$$z_{xy} = \frac{E_x}{H_y} = \frac{E_0 \exp(-ik_z z)}{H_0 \exp(-ik_z z)}$$

$$= \frac{E_0}{H_0}$$

$$= \frac{E_0}{\frac{k_z}{\omega \mu_0} E_0}$$

$$z_{xy} = \frac{\omega \mu_0}{k_z}$$

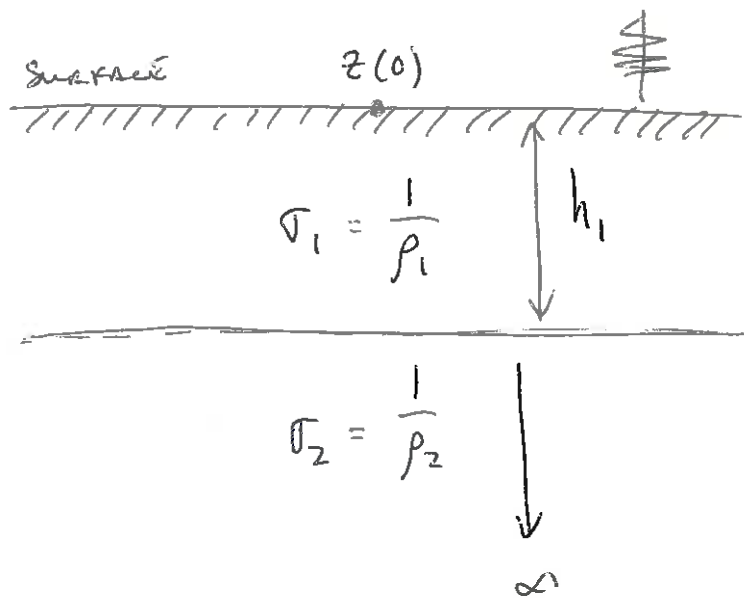
$$= (1+i) \left(\frac{\omega \mu_0}{2\sigma_a} \right)^{1/2}$$

$$= (1+i) \left(\frac{\omega \mu_0 \rho_a}{2} \right)^{1/2}$$

$$\rho_a(\omega) = -\frac{i}{\omega \mu_0} z_{xy}^2$$

$$\rho_a = \frac{1}{\omega \mu_0} |z_{xy}|^2 \quad \text{APPARENT RESISTIVITY}$$

$$z_{xy} = |z_{xy}| e^{i\phi} \quad \phi \quad \text{PHASE ANGLE}$$



$$Z(0) = \frac{\omega \mu_0}{k_1} \left(\frac{k_1 + k_2 \tanh(i k_1 h_1)}{k_2 + k_1 \tanh(i k_1 h_1)} \right)$$

$$\tanh z = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$

$$\tanh(0) = 0$$

$$\lim_{z \rightarrow \infty} \tanh(z) = 1$$

